

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Easy

Question

	Live east of the river	Live west of the river	Total
Less than 40 years old	17	11	28
At least 40 years old	18	89	107
Total	35	100	135

The table summarizes members of a local organization by age and whether they live east or west of the river. If a member of the organization is selected at random, what is the probability that the selected member is at least **40** years old?

Answer

- A. $\frac{28}{135}$
- B. $\frac{35}{135}$
- C. $\frac{100}{135}$
- D. $\frac{107}{135}$

Correct Answer: D

Rationale

Choice D is correct. If a member of the organization is selected at random, the probability that the selected member is at least **40** years old is equal to the number of members who are at least **40** years old divided by the total number of members. According to the table, there are a total of **135** members of the organization, and **107** of these members are at least **40** years old. Therefore, the probability that the selected member is at least **40** years old is $\frac{107}{135}$.

Choice A is incorrect. This is the probability that the selected member is less than **40** years old.

Choice B is incorrect. This is the probability that the selected member lives east of the river.

Choice C is incorrect. This is the probability that the selected member lives west of the river.